

Dietary Advanced Glycation End Products and Aging

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Advanced glycation end products (AGEs) are a heterogeneous, complex group of compounds that are formed when reducing sugar reacts in a non-enzymatic way with amino acids in proteins and other macromolecules. This occurs both exogenously (in food) and endogenously (in humans) with greater concentrations found in older adults. While higher AGEs occur in both healthy older adults and those with chronic diseases, research is progressing to both quantify AGEs in food and in people, and to identify mechanisms that would explain why some human tissues are damaged, and others are not. In the last twenty years, there has been increased evidence that AGEs could be implicated in the development of chronic degenerative diseases of aging, such as cardiovascular disease, Alzheimer's disease and with complications of diabetes mellitus. Results of several studies in animal models and humans show that the restriction of dietary AGEs has positive effects on wound healing, insulin resistance and cardiovascular diseases. Recently, the effect of restriction in AGEs intake has been reported to increase the lifespan in animal models. This paper will summarize the work that has been published for both food AGEs and in vivo AGEs and their relation with aging, as well as provide suggestions for future research.